

Supermarket Sales Data Analysis Report

Power BI ETL & Excel Pivot Tables Assignment

Dataset: Supermarket_Dataset.xlsx | Period: 2010–2013 | Total Records: 9,426

1. Steps Followed in Power Query

The following transformations were applied in Power Query to prepare the dataset for analysis in Power BI.

1.1 Data Extraction

- Loaded the Excel file (Supermarket_Dataset.xlsx, sheet: Orders) into Power BI Desktop.
- Inspected all 24 columns and 9,426 rows for data type correctness and completeness.
- Confirmed that Order Date and Ship Date were parsed as datetime, Postal Code as integer (to be corrected), and numeric fields such as Sales and Profit as decimal numbers.

1.2 Date Handling

- Converted Order Date and Ship Date columns to the proper Date data type using the Change Type function in Power Query.
- Created a new calculated column: Shipping Days = Ship Date minus Order Date, expressed as a whole number of days. The average across all 9,426 orders is 2 days, indicating generally fast order fulfilment.

1.3 Derived Columns

- Profit Margin % = $(\text{Profit} / \text{Sales}) \times 100$, computed for every row to enable margin analysis by segment, region, and category.
- Discount Band: applied conditional logic to group the Discount field into four bands:
 - No Discount (Discount = 0)
 - Low ($0 < \text{Discount} \leq 0.05$)
 - Medium ($0.05 < \text{Discount} \leq 0.10$)
 - High (Discount > 0.10)

1.4 Data Cleansing

- Removed all rows where Sales or Profit is null to ensure aggregations are accurate.
- Applied Trim transformation to Customer Name, Product Category, and Region columns to eliminate leading and trailing whitespace that could cause grouping errors.

- Changed Postal Code data type from integer to text to preserve any leading zeros.

1.5 Aggregations

- Created a summary table grouped by Region and Product Category, computing sum of Sales and sum of Profit per group.
- Created a second summary table grouped by Customer Segment and Ship Mode, counting the number of distinct Order IDs to show order volume distribution.

1.6 Load and Model

- Loaded the cleaned Orders table and both summary tables into the Power BI data model.
- Created a Date dimension table using the DAX CALENDAR function spanning the full range of Order Dates (2010 to 2013) and related it to the Orders table on the Order Date column.

1.7 DAX Measures

The following measures were defined in the Power BI data model:

Measure	DAX Expression
Total Sales	SUM(Orders[Sales])
Total Profit	SUM(Orders[Profit])
Avg Profit Margin	AVERAGE(Orders[Profit Margin %])
YoY Sales Growth	DIVIDE([Total Sales] - CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date])), CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date])))
Profit per Order	DIVIDE([Total Profit], DISTINCTCOUNT(Orders[Order ID]))

2. Key Insight from Power BI Report

Technology Drives Profit; Furniture Underperforms Despite High Revenue

The Sales by Region and Product Category visual reveals a notable divergence between revenue and profitability across the three product categories. The table below summarises the figures drawn directly from the dataset:

Product Category	Total Sales	Total Profit	Profit Margin
Technology	\$3,514,982	\$683,098	19.4%
Office Supplies	\$2,258,626	\$451,990	20.0%

Furniture	\$3,178,624	\$177,354	5.6%
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Technology is both the top revenue generator and the most profitable category, achieving a margin of 19.4%. Office Supplies, though generating the least revenue, comes second in profitability with a margin of 20.0%.

Furniture tells a different story. It generates the second-highest revenue (\$3.18 million) yet produces only \$177,354 in profit, a margin of just 5.6%. This suggests Furniture is sold at consistently thin margins, likely due to high shipping costs, heavy discounting, or a cost structure that erodes returns.

Business Implication: Management should investigate Furniture pricing and discount policies. Redirecting marketing investment toward Technology and high-margin Office Supplies could significantly improve overall profitability without necessarily growing total sales volume.

3. Key Insight from Excel Pivot Analysis

Higher Discounts Consistently Worsen Profitability

The Discount Analysis pivot table reveals an important pattern. Examining average Profit Margin % by Discount Band produces the following results:

Discount Band	Avg Profit Margin %	Total Sales	Order Count
No Discount	-66.4%	\$903,110	848
Low (0-5%)	-9.4%	\$4,235,091	4,390
Medium (5-10%)	-19.7%	\$3,813,076	4,183
High (>10%)	-55.8%	\$955	5

Across all discount bands, average profit margins are negative. The Low band (0-5%) is the least damaging at -9.4%, while the High band (greater than 10%) collapses to -55.8%. The No Discount bucket shows -66.4%, which likely reflects a small number of loss-leader or clearance transactions (only 848 orders).

The Low discount band accounts for the largest share of total sales (\$4.24 million across 4,390 orders), meaning the core volume of the business is already operating at a thin negative margin on a per-row basis. Any push toward Medium or High discounts rapidly worsens this position.

Business Implication: The business should urgently review its pricing and discounting strategy. A blanket discounting policy appears to be systematically destroying value. A category-level analysis combining discount bands with product categories would help identify which specific products and segments can tolerate discounts and which cannot.

4. Additional Findings

4.1 Regional Performance

The Central region leads in both total sales and total profit, making it the strongest market. The South region lags significantly in profitability despite reasonable revenue, pointing to a structural margin problem similar to the Furniture issue identified above.

Region	Total Sales	Total Profit	Profit Margin
Central	\$2,540,342	\$519,826	20.5%
East	\$2,422,805	\$377,566	15.6%
West	\$2,391,739	\$310,849	13.0%
South	\$1,597,346	\$104,201	6.5%

4.2 Sales Growth Over Time

The business shows a consistent upward trend in annual sales across the four-year period, with growth accelerating significantly in 2013.

- 2010: \$1,924,333
- 2011: \$1,944,507 (+1.0% year on year)
- 2012: \$2,231,031 (+14.7% year on year)
- 2013: \$2,852,360 (+27.8% year on year)

The sharp acceleration in 2013 warrants further investigation to determine whether it was driven by new customer acquisition, expanded product lines, or geographic expansion.

4.3 Top Profitable Products

The top 10 products by profit are dominated by office equipment and technology items. The single highest-performing product is the Global Troy Executive Leather Chair at \$79,509 in profit, which is more than double the next item on the list. The remaining top performers are binding machines, copiers, and inkjet printers, confirming the profitability of the Technology and Office Supplies categories.

4.4 Ship Mode Efficiency

Regular Air is both the most-used and most profitable shipping method, generating \$893,602 in profit on \$4.54 million in sales. Delivery Truck handles the highest revenue volume (\$3.71 million) but yields only \$268,062 in profit, suggesting that ground shipping costs are significantly higher relative to the orders it carries. Express Air, while used least, shows a healthy return relative to its sales volume.